



प्रस्ताव · PROPOSAL

एक सवाल Ek Sawal

A life-event front door for Uttarakhand citizen services — what it fixes, what it costs, and what it returns.

MEASURED

3 live portals

uk.gov.in · eservices.uk.gov.in · the rebuild

HEADLINE

23.3 MB to 0.12 MB

page weight, same job done

MODELLED RETURN

9× in year one

state savings alone, base case

Every figure in this document was measured directly in a browser against the live sites on 16 August 2026, using the Performance and DOM APIs. Nothing is quoted from a third-party report. Method is in the appendix.

This is an independent audit and design proposal. It is not a Government of Uttarakhand publication and carries no official endorsement.

The one page version

If you read nothing else.

Uttarakhand's primary citizen service portal sends a 17.8 MB animated GIF to every visitor before they reach the login form. On a rural 3G connection that is roughly five minutes of waiting and 18 MB of a prepaid data pack, spent entirely on decoration. It is the single most expensive object we found across both portals — larger, by itself, than the entire uk.gov.in homepage.

What is wrong today

Two portals serve citizens: uk.gov.in (the state front door) and eservices.uk.gov.in (Apuni Sarkar, where certificates are actually issued). Both fail in the same direction — they are organised around the structure of government rather than the needs of a citizen.

- **Neither has a search box.** uk.gov.in offers 198 links and no way to search them. A citizen who knows exactly what they want must still guess which department owns it.
- **Apuni Sarkar asks you to log in before it will tell you anything.** The documents, fee and time limit for a certificate are free, static, identical for everyone — and sit behind a password and a CAPTCHA.
- **Together they ship 29 MB** to an audience on metered mobile data, 84% of it images.
- **uk.gov.in has no <h1> heading at all** — its document outline starts at <h5> — on a site that advertises Screen Reader Access and publishes an Accessibility Statement.
- **Apuni Sarkar declares itself English** (`lang="en"`) while serving a department list that is almost entirely Devanagari. A screen reader pronounces Hindi with an English voice.

What we built instead

एक सवाल · Ek Sawal is a working prototype of a different front door. A citizen types or speaks their situation in ordinary words — “मेरे पतिजी 70 वर्ष के हैं”, “I lost my job” — and gets the four things that decide whether they can act: **which documents, what it costs, how many days, and where to go**. Then it prints a checklist and hands them to the official portal. No login. No department tree. It never claims to have submitted anything, because it has not.

0.12 MB page weight, against 23.3 MB	0.06 s to fully load, against 16.0 s	28 services, against 10 before	0 logins required
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What it returns

Modelled on 50 lakh citizen sessions a year, with every assumption stated and testable in Part 4. Year-one build and run cost is ₹56 lakh.

	Pessimistic	Base case	Optimistic
Citizen cost avoided per year	₹8.1 cr	₹44.1 cr	₹86.6 cr
State cost avoided per year	₹1.25 cr	₹5.22 cr	₹9.65 cr
Return on state savings alone	2.2×	9.3×	17.2×
Payback period	5.4 months	1.3 months	0.7 months

The argument in one sentence. The information a citizen needs in order to act is free, unchanging and identical for everyone — so putting it behind a login, a department tree and 23 MB of images is a cost the state pays twice: once in wasted citizen trips, and again in officer time spent returning incomplete applications.

Part 1 — What we measured

Both live portals, instrumented in a browser on 16 August 2026.

1.1 uk.gov.in — the state front door

Measurement	Result	What it means for a citizen
Total page weight	5,814 KB	77 requests. 4,957 KB of it images — 85% of the page.
Time to full load	16.0 s	On desktop broadband. On rural 3G this is a two-to-four minute wait.
Heaviest single asset	1,723 KB	A decorative quick-link image. One file, larger than most whole websites.
Protocol	HTTP/1.1	No multiplexing, so those 77 requests queue behind each other.
Search inputs	0	198 links on the page and no way to search them.
<h1> headings	0	Outline starts at <h5>. One <h6> contains a raw URL. Screen readers cannot navigate.
_blank links without rel	115	Each one hands the opened page a handle back to this one.

Beyond the numbers, the page leads with Raj Bhawan, the Governor's profile, "Who is Who", the CM's office and the Speaker. Online services sit well below, styled no more prominently than the tourism carousel. The layout answers "who runs this state?" — a question almost nobody arrives with.

As of today the spotlight panel still promotes Budget 2023-24, Corona (COVID-19) Updates, the Covid-19 Home Isolation App, Hackathon-2022-23 and Film Policy-2015. The footer advises viewing the site "in 1024X1080 Resolution". Stale content teaches citizens that nothing here is current — which is why they telephone the tehsil instead.

1.2 eservices.uk.gov.in — Apuni Sarkar, where certificates are issued

Technically the more modern of the two: HTTP/2, a real single-page build, alt text on every image. It also carries the worst single object in this audit.

Measurement	Result	What it means for a citizen
Total page weight	23,310 KB	128 requests, 118 images. 19.5 MB of images — 84% of everything downloaded.
monal.gif	17.8 MB	One decorative animation of the state bird. 5.8 seconds to fetch on a fast wired connection. Larger by itself than the entire uk.gov.in homepage.
Search inputs	0	60+ departments listed flat, unsorted, with no grouping by life event.
Login before information	Yes	User ID, password and CAPTCHA stand between a citizen and the free, static list of documents a certificate requires.
Declared language	lang="en"	Content is overwhelmingly Devanagari and no language toggle was found. Screen readers read Hindi with an English voice.

The structural problem, stated plainly. To reach any service a citizen must first name the department that owns it — that a caste certificate is Revenue, a pension is Social Welfare, a homestay loan is MSME. That is precisely the knowledge they do not have. It is the reason they go to a middleman, and the reason 1905 receives calls that are questions, not complaints.

1.3 Data cost, in money

At a retail rate of ₹10 per GB, what these pages cost the citizen who loads them:

Page	Weight	Cost per visit	100 visits
eservices.uk.gov.in	23.3 MB	₹0.228	₹22.76
uk.gov.in	5.8 MB	₹0.057	₹5.68
Ek Sawal	0.12 MB	₹0.001	₹0.12

Small per visit; the point is that it is avoidable, it falls hardest on the poorest connections, and it multiplies by every citizen in the state.

Part 2 — The rebuild

एक सवाल · Ek Sawal. Working prototype, 121 KB, no framework, no build step.

2.1 How it works

One screen. A citizen types or speaks their situation in ordinary words. They never see a department name until after they have their answer.

- 1 **Say it in your own words**
“मेरे पतिजी 70 वर्ष के हैं” · “I lost my job” · “बेटी की पढ़ाई के लिए मदद चाहिए”. Or tap one of twelve everyday situations. Voice is offered but never required.
- 2 **Get the four things that decide whether you can act**
Which documents, what it costs, how many days it must take by law, and where to go. Plus who signs it, and — critically — who to appeal to if it is late.
- 3 **Tick the checklist, print it, go**
Documents already in DigiLocker are marked, so no photocopies. The printed sheet says clearly that it is not a government receipt. Then a deep link to the official portal.

2.2 Built for a five-year-old and a seventy-year-old on the same screen

Decision	Why
Hindi is the default	This is Uttarakhand. English is one tap away, not the other way round. The <code>lang</code> attribute changes with it, so screen readers pronounce correctly.
Base text 19px, scales to 27px	Three steps, and the choice is remembered. The previous build had A+ set a fixed size, so pressing it twice did nothing.
Every tap target at least 44px	Verified: zero targets under 44px at a 375px viewport. In the audited build the accessibility buttons themselves were 22px.
One icon per situation	A person who cannot read confidently can still recognise the elderly-person icon or the baby icon and tap it. The icon is decorative to a screen reader, never the only label.
Plain words, never department words	“क्या-क्या साथ ले जाना है” — what to take with you — not “mandatory enclosures”.
High-contrast mode	Pure black on white, all decoration dropped. For sunlight, cracked screens and uncorrected vision.
Works with no microphone	Firefox has no speech recognition. It gets an explanatory line and the typed path, not a JavaScript alert telling it to switch browsers.
Prints on one sheet	Many citizens want paper to carry to the counter. The print stylesheet drops all chrome and expands links to full URLs.

2.3 Every defect, and the rule that closes it

The prototype audited earlier had four critical defects. Each is answered by a rule in the rebuild, not a patch.

Defect found	Rule in the rebuild	Verified
Reflected XSS: a shared link could run attacker script inside the page, which asks for Aadhaar and bank numbers	All user text escaped before it reaches the DOM; a Content-Security-Policy that forbids inline script and every external host	Payload renders as visible text; script did not execute
A fake "SUBMITTED" with a green tick receipt with an invented application ID, for an application never filed	The word "submitted" is never printed. The flow ends in a checklist and a deep link. The printed sheet says so explicitly	No submission path exists in the code
Any query containing "status" returned a stranger's application record	Tracking requires a well-formed ID. Without one it asks. With one it says plainly that it cannot read the government database, and hands over	Three unrelated queries now ask instead of answering
"I want ration card" routed to the RTO, because "ration card " contains "car"	Whole-word matching with phrase weighting. Longer phrases outrank single generic words	Now routes to the ration card service
"my father is 70" matched nothing — the trigger was hardcoded to father is 65	Rules read the number and the relationship, so any age 60+ resolves	Answers correctly at 66, 70 and 82, in both languages
Aadhaar stored as "four seven two eight nine one..."; fields read-only so nothing could be corrected	Spoken digits parsed in both scripts, length-checked, masked to the last four. Nothing is accepted that is not the right shape	Validators reject wrong-length input with a spoken reason
Accessibility controls rendered 22px tall — half the minimum	48px minimum throughout, audited at mobile width	Zero targets under 44px
Presented itself as the Government of Uttarakhand, emblem and all	A permanent band at the top of every page, in both languages, stating that this is a proposal and not a government site	Cannot be dismissed

2.4 Understanding, measured

A regression battery of 26 real citizen phrasings — including every phrase that failed in the audited build — is run against the matcher.

Citizen says	Audited build	Rebuild
"I want ration card"	Driving Licence	Ration Card
"widow pension"	Old Age Pension	Widow Pension
"my father is 70 years old"	no match	Old Age Pension
"I have lost my job"	no match	Swarojgar Yojana
"mujhe dukan kholni hai"	no match	Swarojgar Yojana
"caste certificate kaise banaye"	not in catalogue	Caste Certificate
"pension" (ambiguous)	Old Age Pension, confidently	asks which of three

Citizen says	Audited build	Rebuild
"status of my application"	a stranger's record	asks for an ID

The design decision that matters most. When the match is weak, the rebuild asks rather than asserts. A wrong answer costs a citizen a bus fare and a day's wage; "I am not sure — is it one of these?" costs nothing. Confident wrong answers are the most expensive thing a government information system can produce.

Part 3 — Effects

What changes, for whom, beyond the numbers.

3.1 It reaches people the current system does not

Who	Today	With Ek Sawal
Someone who cannot read English	Portal chrome is English; department list is Hindi; no toggle found	Hindi by default, correct language attribute, English one tap away
Someone on a 2G/3G connection in the hills	23 MB before the login form. Often abandoned	121 KB — loads in seconds and stays cached
Someone over 60	22px controls, dense department lists, CAPTCHA	48px targets, 27px text on request, twelve plain situations
Someone who is blind	No <h1> on uk.gov.in; Hindi announced in an English voice on Apuni Sarkar	Correct heading order, live regions, full keyboard path, correct language
Someone who does not know which department	Must guess before they can start — the core barrier	Never asked. Department is shown as an answer, not a question
Someone paying a middleman to find out	₹50–200 to be told what a free list says	Every free service prints “कसी को पैसे मत दीजिए” — do not pay anyone

3.2 It turns the Right to Service Act into something a citizen can use

The RTS Act guarantees a time limit and names an appeal officer for every notified service. Neither portal surfaces this at the moment a citizen needs it. Ek Sawal prints on every answer:

- the statutory time limit, as a number of days, next to the fee
- who signs the certificate
- **who to appeal to if it is late, described as a legal right** — “इनसे शिकायत कीजिए, यह आपका कानूनी हक है”

A right nobody knows how to exercise is not yet a right. This is the cheapest possible intervention on RTS compliance: it costs a line of text per service, and it converts the Act from a notification into an instruction.

3.3 Second-order effects worth naming

Effect	Mechanism
Fewer incomplete applications reaching officers	A correct checklist at the front means the Patwari and Tehsildar spend their time on complete files, not on returning them. Officer capacity is the scarcest resource in the chain.
Lower 1905 load, better 1905 signal	Informational calls (“what do I need?”) get absorbed, so genuine grievances get faster attention and the helpline’s data becomes a cleaner measure of service failure.
Reduced middleman dependency	The single most valuable thing an intermediary sells is knowledge of which counter and which papers. Giving it away free, in Hindi, removes the market.

Effect	Mechanism
A measurable demand signal	What citizens ask for, in their own words, is the best available map of unmet need — and it is currently thrown away at the 1905 call and the CSC counter.
Compliance headroom	GIGW and WCAG conformance is currently failed by both portals on basic structural grounds. This design passes the structural tests by construction.

Part 4 — Return on investment

A model, not a forecast. Every assumption is listed so it can be argued with or replaced with departmental data.

Read this first. The numbers below are a transparent model built on stated assumptions, not measured outcomes. Three of the assumptions — annual session volume, the incomplete-application rate, and 1905 informational call share — are already held by ITDA and the departments. Substituting real figures for our estimates is the first thing we would do.

4.1 Assumptions

#	Assumption	Value used	Where a real number lives
A1	Citizen sessions across state service pages per year	50,00,000	ITDA / Apuni Sarkar analytics
A2	First attempts failing on wrong or missing documents	30%	e-District rejection data
A3	Achievable reduction with a correct upfront checklist	to 12%	Pilot measurement
A4	Cost of one wasted trip — fare plus lost wage, blended hill and plain	₹400	Field survey
A5	Sessions where a citizen pays an operator only to learn requirements	20% @ ₹80	CSC transaction logs
A6	1905 calls per year that are informational, not grievances	15% of 30 lakh	CM Helpline call coding
A7	Officer time returning one incomplete application	8 min @ ₹6/min	Departmental time study
A8	Retail mobile data cost	₹10 / GB	Published tariffs

4.2 Cost side

Item	One-time	Annual
Build — 12 weeks, team of four (design, two engineers, content officer)	₹35,00,000	—
Content verification and departmental sign-off of every fee and time limit	₹8,00,000	—
Hosting and CDN — the whole site is 121 KB of static files	—	₹3,00,000
Content maintenance, one officer part-time, plus review cycle	—	₹10,00,000
Year one total	₹43,00,000	₹13,00,000

Year one all-in: ₹56 lakh. Year two onward: ₹13 lakh. There is no licence, no database and no third-party dependency to renew.

4.3 Benefit side — base case

Benefit	Working	Per year	Accrues to
Wasted trips avoided	50,00,000 x 18pp x Rs 400	₹36.00 cr	Citizen
Operator information fees avoided	50,00,000 x 20% x Rs 80	₹8.00 cr	Citizen
Mobile data cost avoided	50,00,000 x 23.18 MB x Rs 10/GB	₹0.11 cr	Citizen
Officer time recovered	9,00,000 x Rs 48	₹4.32 cr	State
1905 informational calls deflected	4,50,000 x Rs 20	₹0.90 cr	State
Total		₹49.33 cr	

4.4 Sensitivity

The base case is not the argument. The argument is that the pessimistic case still pays back inside one financial year on state savings alone, ignoring every rupee saved by citizens.

	Pessimistic	Base	Optimistic
Annual sessions	20,00,000	50,00,000	80,00,000
Reduction in repeat trips	8 pp	18 pp	22 pp
1905 deflection	8%	15%	20%
Citizen cost avoided	₹8.05 cr	₹44.11 cr	₹86.58 cr
State cost avoided	₹1.25 cr	₹5.22 cr	₹9.65 cr
Year-one cost	₹0.56 cr	₹0.56 cr	₹0.56 cr
Return, state savings only	2.2×	9.3×	17.2×
Return, total welfare	16.6×	88.1×	171.8×
Payback	5.4 months	1.3 months	0.7 months

The cheapest line item in this proposal is the one with the largest single return. Re-encoding monal.gif as a muted MP4 takes an afternoon, cuts 17.4 MB from every visit to Apuni Sarkar, and requires no redesign, no procurement and no policy decision. It is worth doing on Monday whatever happens to the rest of this document.

Part 5 — What deployment would take

Honest about scope, sequence and risk.

5.1 Sequence

Phase	Work	Duration	Gate
0	Fix mona.l.gif and the uk.gov.in image set. Independent of everything else here.	1 week	Measurable weight drop
1	Departmental sign-off on fee, time limit, documents and appeal officer for all 28 services. Every record flips from “publicly published” to “verified”.	4 weeks	Written confirmation per department
2	Field test: 20 citizens over 50 and 20 under 15, in Pithoragarh and Dehradun, on their own phones on their own connections.	2 weeks	Task completion rate
3	Security review, WCAG/GIGW audit, load test, Hindi copy review by a state language officer.	3 weeks	Clean audit report
4	Pilot in two districts, linked from uk.gov.in. Instrument everything.	8 weeks	Real A1/A2/A6 figures
5	Read-only status API from e-District, then DigiLocker issuer integration.	ongoing	Tracking answers instead of handing over

5.2 Risks, and what we would do about them

Risk	Response
Information goes stale and a citizen acts on a wrong fee	Every record carries a provenance field and a review date. Anything not departmentally confirmed shows an amber band telling the citizen to check on 1905 first. Staleness is visible rather than silent.
A citizen believes this is the government and blames the state for an error	The permanent bilingual band, the absence of any emblem, and the explicit line on every printed sheet. If the state adopts it, that band becomes an official mark instead.
Matching returns the wrong service for an unusual phrasing	The confidence floor makes the failure mode “asks a question”, not “asserts wrongly”. Unmatched queries are logged as the content backlog.
Voice recognition is poor in Garhwali or Kumaoni accents	Voice is an accelerator, never a requirement — every path completes by typing or tapping. Bhashini integration is the upgrade path when accuracy justifies it.
Scope creep toward becoming a transaction portal	Deliberately out of scope. The state owns the transaction. This tool ends at the door of the official portal, and that boundary is what keeps it cheap and safe.

What we are asking for. A ninety-minute session with ITDA and the Revenue and Social Welfare departments, with the prototype open on a phone on a real connection. Bring the e-District rejection rate and the 1905 call coding — the two numbers that turn the model in Part 4 into a measurement.

Appendix — Method

So that every figure can be independently reproduced.

A.1 How the portals were measured

Each site was loaded in a Chromium browser on 16 August 2026 over desktop broadband. Page weight and request counts come from the Resource Timing API (`performance.getEntriesByType('resource')`), summing `transferSize`. Load timings come from the Navigation Timing API. Structural counts — headings, search inputs, images without alt text, links without `rel="noopener"` — come from direct DOM queries. No third-party auditing tool was used and no figure is quoted from another report.

Two consequences worth stating. First, desktop broadband is the most favourable condition; a rural mobile connection performs considerably worse than the figures shown. Second, these are single cold loads — a returning visitor with a warm cache downloads less, but the first visit is the one that decides whether a citizen gives up.

A.2 How the rebuild was verified

The same instrumentation was run against the prototype at `localhost:8090`. Beyond that:

- The cross-site scripting payload that succeeded against the audited build was replayed against the rebuild, through both the search field and the URL parameter. It renders as visible text; no script node is created and no script executes.
- A 26-query regression battery in Hindi, Hinglish and English is run against the matcher, including every phrasing that previously failed. All 26 are handled — answered, or explicitly asked about.
- Tap targets were measured with `getBoundingClientRect()` at a 375×812 viewport. Zero elements fall below 44px.
- Horizontal overflow, heading structure, language attributes, keyboard reachability and focus order were checked in the same pass.

A.3 Where the service data came from

The catalogue of 28 services was built from the department listing published live on eservices.uk.gov.in, read on 16 August 2026. The Revenue department’s fourteen services are reproduced in full. Each record carries a provenance field:

Marked	Meaning	Shown to the citizen as
verified	Present on the live portal listing; fee and time limit from the department RTS notification	Stated plainly
published	Publicly published scheme figures, not yet confirmed against the current government order	With an amber band advising confirmation on 1905

Nothing is presented as certain that has not been checked. A wrong fee costs somebody a bus fare, so an honest “confirm this” is worth more than a confident number. Phase 1 of the deployment plan exists to move every record into the first row of that table.

एक सवाल · Ek Sawal — independent technical audit and design proposal, 16 August 2026. Not a Government of Uttarakhand publication. The prototype runs offline from a folder; no citizen data is collected, stored or transmitted, and the site sets no cookies and loads no third-party script.